

InsightGen Solutions – Domain-Specific AI Assistant Optimization Report

1. Executive Summary

This project transformed a general-purpose FLAN-T5 model into a healthcare-focused assistant using decoding optimization, systematic evaluation, and LoRA fine-tuning. Through controlled decoding parameters and PEFT adaptation, we improved factual consistency, domain terminology accuracy, and reliability while maintaining low GPU and memory costs.

2. Decoding Optimization

- Greedy Decoding (temperature=0): Best for deterministic, factual responses such as medical explanations.
- High Creativity (temperature=1.2, top-p=0.95): More diverse outputs but increased hallucination risk.
- Controlled Creativity (temperature=0.7, top-k=50): Balanced reliability and clarity.
- Beam Search (num_beams=4): Improved coherence for longer structured responses.
- Optimal Choice: Controlled Creativity (temperature=0.7, top-p=0.9) for production healthcare use.

3. Evaluation Framework

- Automated Metrics: BLEU, ROUGE-L, BERTScore (semantic similarity).
- Qualitative Ratings: Coherence (1–5), Factuality (1–5), Relevance (1–5).
- Finding: BERTScore correlated better with human judgment than BLEU/ROUGE.
- Blended Strategy: Use automated metrics for screening + human review for high-risk medical outputs.

4. PEFT (LoRA) Fine-Tuning

- Base Model: google/flan-t5-small.
- LoRA Config: r=8, alpha=16, target_modules=['q', 'v'], dropout=0.1.
- Trainable Parameters Reduced by >99% compared to full fine-tuning.
- Adapter Size: ~5MB vs full model hundreds of MB.

- Training Time: Under 20 minutes on standard GPU.
- Result: Improved domain terminology accuracy and reduced hallucinations.

5. Cost and Trade-Off Analysis

- LoRA provided strong quality gains with minimal GPU cost.
- Full fine-tuning would require 10–100x more memory and time.
- Decoding optimization provided fastest performance improvement.
- Best cost-effective strategy: Combine decoding control + LoRA adaptation.

6. Production Recommendation

- Model: FLAN-T5 + LoRA adapter.
- Decoding: temperature=0.7, top-p=0.9, num_beams=4 for structured tasks.
- Monitoring: Track BERTScore trend, hallucination flags, response latency.
- Retraining: Quarterly updates with new domain examples.
- Risk Mitigation: Human oversight for critical healthcare decisions.